

example poles: $0, -2$

angle test: $-\angle s - \angle(s + 2) = (2k + 1)\pi$

if false: point is not on root locus

if true: $K^* = |s| |s + 2|$

examples: $-0.4 \mapsto 0.64, -1 \mapsto 1, -1 + 2j \mapsto 5$